

Qualification Pack

Artificial Intelligence Associate

QP Code: NIE/SSC/Q1004

Version: 1.0

NSQF Level: 4

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Qualification Pack

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NIE/SSC/Q1004: Artificial Intelligence Associate

Brief Job Description

An Artificial Intelligence Associate collaborates with AI experts to develop, implement, and optimize AI algorithms and models, leveraging data analysis and programming skills to drive innovation and solve complex problems across various domains.

Personal Attributes

An Artificial Intelligence Associate assists in developing and implementing AI solutions, requiring strong analytical skills, programming proficiency, and a deep understanding of machine learning algorithms to contribute effectively to AI projects and initiatives.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ITS/N1005: PROGRAMMING with PYTHON](#)
2. [NIE/ITS/N1006: CONCEPTUALISING DATA SCIENCE with PYTHON](#)
3. [NIE/ITS/N1007: DATA ANALYSIS and VISUALIZATION](#)
4. [NIE/ITS/N1008: FUNDAMENTALS of MACHINE LEARNING](#)
5. [NIE/ITS/N1009: PERFORMANCE and ACCURACY of MACHINE LEARNING MODELS](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Artificial Intelligence
Occupation	AI Development
Country	India
NSQF Level	4
Credits	15

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Aligned to NCO/ISCO/ISIC Code	2511.0106: Support Engineer
Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 OR 10th grade pass and pursuing continuous schooling OR 10th grade pass with 2 Years of experience OR Previous relevant Qualification of NSQF Level (3.0 with minimum education as 8th Grade pass) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-04-IT-00357-2023-V1-NIELIT
NQR Version	1

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NIE/ITS/N1005: Development of Deep Learning framework for collecting, analyzing and interpreting large amounts of data

Description

This NOS delves into creating deep learning frameworks tailored for collecting, analyzing, and interpreting vast data sets, empowering students to harness the potential of deep learning for comprehensive data insights and decision-making in big data scenarios.

Scope

The scope covers the following :

- Model Architecture and Optimization, Distributed Computing and Parallel Processing

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Engineer Deep Learning architectures using frameworks like TensorFlow or PyTorch that cater to the complexities of large datasets. Experiment with convolutional neural networks (CNNs) and transformer-based models to capture intricate patterns.
- PC2.** Ability to develop techniques for distributing the training process across multiple GPUs or even across a cluster of machines.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
 - A.Organizational Context
 - KA1. Efficient Data Handling
 - KA2. Insightful Analysis
 - KA3. Scalable Architecture
- KU2.**
 - B. Technical Knowledge
 - KB1.Data Processing and Pre processing
 - KB2.Deep Learning Architectures
 - KB3.Parallel Computing and Optimization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
 - SA1.Fill up documentation applicable to ones role.
 - SA2. Read English and/or vernacular language.
 - SA3. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA4. Ability to read from different sources- books screens in machines and signage.
 - SA5. Understand the various color codes, as per standard electrical,mechanical
 - SA6. Express statements or information clearly so that others can hear and understand.

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- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - The user/individual on the job needs to know and understand how to :
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	24	-	6
PC1. Engineer Deep Learning architectures using frameworks like TensorFlow or PyTorch that cater to the complexities of large datasets. Experiment with convolutional neural networks (CNNs) and transformer-based models to capture intricate patterns.	20	12	-	3
PC2. Ability to develop techniques for distributing the training process across multiple GPUs or even across a cluster of machines.	20	12	-	3
NOS Total	40	24	-	6

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1005
NOS Name	Development of Deep Learning framework for collecting, analyzing and interpreting large amounts of data
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence , Data Science & Engineering
Occupation	AI Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1005: PROGRAMMING with PYTHON

Description

Python Programming cover the installation and configuration of a Python environment, basic programming concepts, data structures, and the use of modules and packages to provide a foundational understanding of Python development.

Scope

The scope covers the following :

- IDE Installation, Programming, Exploring Data Structures

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming
- PC2.** Understand the basics of Python Language and will be able to recognize Python syntax
- PC3.** Understand and write programs in python language, compile, debug and handle exceptions
- PC4.** Working with top down or bottom up approach by making functions.
- PC5.** Do modular programming by defining and calling various functions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Python Environment Setup: The student will be able to set up a Python programming environment, including installing Python and configuring essential tools like Integrated Development Environments (IDEs) or code editors.
- KU2.** Basic Programming Skills: The student will have a solid grasp of fundamental programming concepts, such as variables, data types, operators, conditional statements, loops, and functions.
- KU3.** Datatypes and Operators: They will understand how to work with various data types (e.g., integers, strings, lists) and apply operators (e.g., arithmetic, comparison, logical) to manipulate data.
- KU4.** Control Flow: The student will be able to create programs with control flow structures, including conditional statements (if, else, and elif) and loops (for and while) to control program execution.
- KU5.** Functions: They will know how to define and use functions, including passing arguments, returning values, and organizing code for reusability.
- KU6.** Data Structures: The student will gain knowledge about different data structures like lists, dictionaries, sets, and tuples, and understand how to select the appropriate data structure for specific tasks.

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KU7. Modules and Packages: They will learn how to create, import, and use Python modules and packages to organize and manage code efficiently, promoting code modularity and reusability.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Students will develop general skills like problem-solving, logical thinking, attention to detail, and communication.
- GS2.** Students will also acquire professional skills, including programming proficiency, code efficiency, modular programming, adaptability, and knowledge of software development best practices.
- GS3.** This comprehensive skill set equips students for diverse career opportunities in technology and programming, providing a strong foundation for their professional development

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	28	17	-	-
PC1. Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming	5	2	-	-
PC2. Understand the basics of Python Language and will be able to recognize Python syntax	8	4	-	-
PC3. Understand and write programs in python language, compile, debug and handle exceptions	5	4	-	-
PC4. Working with top down or bottom up approach by making functions.	5	3	-	-
PC5. Do modular programming by defining and calling various functions	5	4	-	-
NOS Total	28	17	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1005
NOS Name	PROGRAMMING with PYTHON
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	2
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1006: Create and deploy interactive Artificial Intelligence projects

Description

In this NOS, participants will learn to design and deploy interactive Artificial Intelligence projects, gaining practical skills to develop AI applications and bring them to life, fostering hands-on experience in AI project implementation.

Scope

The scope covers the following :

- Project Ideation and Design, Model Development and Integration

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Ability to develop innovative AI project ideas that address real-world challenges or provide valuable experiences and design user-centered interfaces that enable seamless interaction with AI functionalities.
- PC2.** Develop AI models using Deep Learning techniques tailored to your project's goals. Integrate AI capabilities such as natural language processing, computer vision, or recommendation systems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
 - A.Organizational Context
 - KA1. Enhanced User Engagement
 - KA2.Data-Driven Decision Making
 - KA3. Innovation and Differentiation
- KU2.**
 - B. Technical Knowledge
 - KB1.AI Algorithms and Models
 - KB2.Deployment and APIs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
 - SA1.Fill up documentation applicable to ones role.
 - SA2. Read English and/or vernacular language.
 - SA3. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA4. Ability to read from different sources- books screens in machines and signage.
 - SA5. Understand the various color codes, as per standard electrical,mechanical
 - SA6. Express statements or information clearly so that others can hear and understand.

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- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - The user/individual on the job needs to know and understand how to :
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	24	-	6
PC1. Ability to develop innovative AI project ideas that address real-world challenges or provide valuable experiences and design user-centered interfaces that enable seamless interaction with AI functionalities.	20	12	-	3
PC2. Develop AI models using Deep Learning techniques tailored to your project's goals. Integrate AI capabilities such as natural language processing, computer vision, or recommendation systems.	20	12	-	3
NOS Total	40	24	-	6

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1006
NOS Name	Create and deploy interactive Artificial Intelligence projects
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence , Data Science & Engineering
Occupation	AI Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1006: CONCEPTUALISING DATA SCIENCE with PYTHON

Description

This NOS provides a comprehensive introduction to data science concepts and techniques using the Python programming language. Students learn how to analyze data, build predictive models, and gain a foundational understanding of data science principles.

Scope

The scope covers the following :

- Data Analysis and Visualization, Machine Learning Fundamentals, Conceptual Understanding of Data Science

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Students should be able to effectively explore and analyze datasets using Python, including data cleaning, feature engineering, and visualization.
- PC2.** Students should exhibit a solid grasp of the foundational concepts in data science, such as statistical methods, data ethics, and the business implications of data analysis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Python Programming: Students should acquire a strong understanding of Python programming, including data structures, control flow, and the use of libraries such as NumPy and Pandas for data manipulation.
- KU2.** Data Analysis Techniques: A solid foundation in data analysis principles, including data cleaning, transformation, and exploratory data analysis (EDA), is crucial. Students should be knowledgeable about various statistical and visualization techniques used to derive insights from data.
- KU3.** Data Science Concepts: Beyond the technical aspects, students should gain a conceptual understanding of data science, encompassing ethical considerations, the data science lifecycle, and practical applications of data science in diverse fields, such as healthcare, finance, and marketing.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.**
- SA1. Fill up documentation applicable to ones role.
 - SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA3. Express statements or information clearly so that others can hear and understand.
 - SA4. Participate in and understand the main points of simple discussions.
- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	28	16	10	-
PC1. Students should be able to effectively explore and analyze datasets using Python, including data cleaning, feature engineering, and visualization.	14	8	5	-
PC2. Students should exhibit a solid grasp of the foundational concepts in data science, such as statistical methods, data ethics, and the business implications of data analysis.	14	8	5	-
NOS Total	28	16	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1006
NOS Name	CONCEPTUALISING DATA SCIENCE with PYTHON
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	3
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1007: Conceptualising Data Science with Python

Description

This NOS provides a comprehensive introduction to data science concepts and techniques using the Python programming language. Students learn how to analyze data, build predictive models, and gain a foundational understanding of data science principles.

Scope

The scope covers the following :

- Data Analysis and Visualization, Machine Learning Fundamentals, Conceptual Understanding of Data Science

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Students should be able to effectively explore and analyze datasets using Python, including data cleaning, feature engineering, and visualization.
- PC2.** Students should exhibit a solid grasp of the foundational concepts in data science, such as statistical methods, data ethics, and the business implications of data analysis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Python Programming: Students should acquire a strong understanding of Python programming, including data structures, control flow, and the use of libraries such as NumPy and Pandas for data manipulation.
- KU2.** Data Analysis Techniques: A solid foundation in data analysis principles, including data cleaning, transformation, and exploratory data analysis (EDA), is crucial. Students should be knowledgeable about various statistical and visualization techniques used to derive insights from data.
- KU3.** Data Science Concepts: Beyond the technical aspects, students should gain a conceptual understanding of data science, encompassing ethical considerations, the data science lifecycle, and practical applications of data science in diverse fields, such as healthcare, finance, and marketing.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.**
- SA1. Fill up documentation applicable to ones role.
 - SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA3. Express statements or information clearly so that others can hear and understand.
 - SA4. Participate in and understand the main points of simple discussions.
- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	29	13	10	-
PC1. Students should be able to effectively explore and analyze datasets using Python, including data cleaning, feature engineering, and visualization.	14	6	5	-
PC2. Students should exhibit a solid grasp of the foundational concepts in data science, such as statistical methods, data ethics, and the business implications of data analysis.	15	7	5	-
NOS Total	29	13	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1007
NOS Name	Conceptualising Data Science with Python
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1007: DATA ANALYSIS and VISUALIZATION

Description

This NOS course provides students with the skills to systematically examine and interpret data through statistical analysis and visualization techniques. Participants learn how to extract meaningful insights, communicate findings effectively, and make data-driven decisions.

Scope

The scope covers the following :

- Data Cleaning and Preparation, Exploratory Data Analysis (EDA), Statistical Analysis, Data Visualization

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Students should demonstrate the ability to clean and preprocess data effectively, removing or handling missing values and outliers, ensuring that data is ready for analysis.
- PC2.** Performance should be measured based on students' skills in conducting EDA, including their ability to generate meaningful visualizations, compute descriptive statistics, and identify patterns and trends in the data.
- PC3.** Performance criteria should include the ability to create clear and informative data visualizations that effectively communicate insights from the data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students should have a deep understanding of data cleaning techniques, data imputation, and the ability to prepare data for analysis by handling missing values, outliers, and inconsistencies.
- KU2.** Knowledge of various EDA techniques, including summary statistics, data visualization methods, and hypothesis testing, is crucial. Students should understand how to use these tools to gain insights from data.
- KU3.** A solid foundation in statistical concepts, such as probability, sampling, and hypothesis testing, is essential. Students should comprehend the application of statistical tests and methods to draw meaningful conclusions from data.
- KU4.** Students should be knowledgeable about data visualization best practices, including understanding chart types, color choices, labeling, and storytelling through data visualization.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.**
- SA1. Fill up documentation applicable to ones role.
 - SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA3. Express statements or information clearly so that others can hear and understand.
 - SA4. Participate in and understand the main points of simple discussions.
- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	42	25	-	-
PC1. Students should demonstrate the ability to clean and preprocess data effectively, removing or handling missing values and outliers, ensuring that data is ready for analysis.	14	8	-	-
PC2. Performance should be measured based on students' skills in conducting EDA, including their ability to generate meaningful visualizations, compute descriptive statistics, and identify patterns and trends in the data.	14	8	-	-
PC3. Performance criteria should include the ability to create clear and informative data visualizations that effectively communicate insights from the data.	14	9	-	-
NOS Total	42	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1007
NOS Name	DATA ANALYSIS and VISUALIZATION
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	3
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1008: Data analysis and Visualization

Description

This NOS course provides students with the skills to systematically examine and interpret data through statistical analysis and visualization techniques. Participants learn how to extract meaningful insights, communicate findings effectively, and make data-driven decisions.

Scope

The scope covers the following :

- Data Cleaning and Preparation, Exploratory Data Analysis (EDA), Statistical Analysis, Data Visualization

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Students should demonstrate the ability to clean and preprocess data effectively, removing or handling missing values and outliers, ensuring that data is ready for analysis.
- PC2.** Performance should be measured based on students' skills in conducting EDA, including their ability to generate meaningful visualizations, compute descriptive statistics, and identify patterns and trends in the data.
- PC3.** Performance criteria should include the ability to create clear and informative data visualizations that effectively communicate insights from the data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students should have a deep understanding of data cleaning techniques, data imputation, and the ability to prepare data for analysis by handling missing values, outliers, and inconsistencies.
- KU2.** Knowledge of various EDA techniques, including summary statistics, data visualization methods, and hypothesis testing, is crucial. Students should understand how to use these tools to gain insights from data.
- KU3.** A solid foundation in statistical concepts, such as probability, sampling, and hypothesis testing, is essential. Students should comprehend the application of statistical tests and methods to draw meaningful conclusions from data.
- KU4.** Students should be knowledgeable about data visualization best practices, including understanding chart types, color choices, labeling, and storytelling through data visualization.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.**
- SA1. Fill up documentation applicable to ones role.
 - SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA3. Express statements or information clearly so that others can hear and understand.
 - SA4. Participate in and understand the main points of simple discussions.
- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	43	21	-	-
PC1. Students should demonstrate the ability to clean and preprocess data effectively, removing or handling missing values and outliers, ensuring that data is ready for analysis.	13	7	-	-
PC2. Performance should be measured based on students' skills in conducting EDA, including their ability to generate meaningful visualizations, compute descriptive statistics, and identify patterns and trends in the data.	14	7	-	-
PC3. Performance criteria should include the ability to create clear and informative data visualizations that effectively communicate insights from the data.	16	7	-	-
NOS Total	43	21	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1008
NOS Name	Data analysis and Visualization
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

Qualification Pack

NIE/ITS/N1008: FUNDAMENTALS of MACHINE LEARNING

Description

This NOS provides a foundational understanding of machine learning concepts, algorithms, and applications, with a focus on teaching the fundamental principles of how machines can learn from data to make predictions and decisions.

Scope

The scope covers the following :

- Supervised Learning, Unsupervised Learning, Model Evaluation and Optimization

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Students should be able to build machine learning models for various tasks, select appropriate algorithms, and evaluate model performance
- PC2.** Performance should be measured based on the students' ability to fine-tune model hyperparameters effectively to optimize model performance and generalization.
- PC3.** Students should demonstrate their problem-solving skills by applying machine learning techniques to real-world problems, interpreting model results, and making informed decisions based on the insights derived from the data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students should have a solid grasp of fundamental machine learning concepts, including supervised and unsupervised learning, feature engineering, model training, and the bias-variance trade-off.
- KU2.** Knowledge of various machine learning algorithms and techniques, such as linear regression, decision trees, k-means clustering, and neural networks, is essential. Students should understand when and how to apply these algorithms.
- KU3.** Understanding of model evaluation methods, cross-validation, hyperparameter tuning, and techniques for preventing overfitting or underfitting is critical for building effective machine learning models.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.**
- SA1. Fill up documentation applicable to ones role.
 - SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA3. Express statements or information clearly so that others can hear and understand.
 - SA4. Participate in and understand the main points of simple discussions.
- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	25	8	-	-
PC1. Students should be able to build machine learning models for various tasks, select appropriate algorithms, and evaluate model performance	8	2	-	-
PC2. Performance should be measured based on the students' ability to fine-tune model hyperparameters effectively to optimize model performance and generalization.	8	3	-	-
PC3. Students should demonstrate their problem-solving skills by applying machine learning techniques to real-world problems, interpreting model results, and making informed decisions based on the insights derived from the data.	9	3	-	-
NOS Total	25	8	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1008
NOS Name	FUNDAMENTALS of MACHINE LEARNING
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	1
Version	1.0
Next Review Date	NA

Qualification Pack

NIE/ITS/N1009: Programming with PYTHON

Description

Python Programming cover the installation and configuration of a Python environment, basic programming concepts, data structures, and the use of modules and packages to provide a foundational understanding of Python development.

Scope

The scope covers the following :

- IDE Installation, Programming, Exploring Data Structures

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming
- PC2.** Understand the basics of Python Language and will be able to recognize Python syntax
- PC3.** Understand and write programs in python language, compile, debug and handle exceptions
- PC4.** Working with top down or bottom up approach by making functions.
- PC5.** Do modular programming by defining and calling various functions
- PC6.** Working with different data structures which allow to organize and store data
- PC7.** Apply algorithms to process data in various data structures in a meaningful way

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Python Environment Setup: The student will be able to set up a Python programming environment, including installing Python and configuring essential tools like Integrated Development Environments (IDEs) or code editors.
- KU2.** Basic Programming Skills: The student will have a solid grasp of fundamental programming concepts, such as variables, data types, operators, conditional statements, loops, and functions.
- KU3.** Datatypes and Operators: They will understand how to work with various data types (e.g., integers, strings, lists) and apply operators (e.g., arithmetic, comparison, logical) to manipulate data.
- KU4.** Control Flow: The student will be able to create programs with control flow structures, including conditional statements (if, else, and elif) and loops (for and while) to control program execution.
- KU5.** Functions: They will know how to define and use functions, including passing arguments, returning values, and organizing code for reusability.

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- KU6.** Data Structures: The student will gain knowledge about different data structures like lists, dictionaries, sets, and tuples, and understand how to select the appropriate data structure for specific tasks.
- KU7.** Modules and Packages: They will learn how to create, import, and use Python modules and packages to organize and manage code efficiently, promoting code modularity and reusability.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Students will develop general skills like problem-solving, logical thinking, attention to detail, and communication.
- GS2.** Students will also acquire professional skills, including programming proficiency, code efficiency, modular programming, adaptability, and knowledge of software development best practices.
- GS3.** This comprehensive skill set equips students for diverse career opportunities in technology and programming, providing a strong foundation for their professional development

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	29	14	-	-
PC1. Learn to install and configure the python IDE and learn about working on the collaborative cloud interface required for programming	5	2	-	-
PC2. Understand the basics of Python Language and will be able to recognize Python syntax	3	2	-	-
PC3. Understand and write programs in python language, compile, debug and handle exceptions	5	2	-	-
PC4. Working with top down or bottom up approach by making functions.	5	2	-	-
PC5. Do modular programming by defining and calling various functions	5	2	-	-
PC6. Working with different data structures which allow to organize and store data	3	2	-	-
PC7. Apply algorithms to process data in various data structures in a meaningful way	3	2	-	-
NOS Total	29	14	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1009
NOS Name	Programming with PYTHON
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ITS/N1009: PERFORMANCE and ACCURACY of MACHINE LEARNING MODELS

Description

This NOS focuses on evaluating and improving the effectiveness of machine learning models by teaching students to use various evaluation metrics and fine-tuning techniques. It covers methods for optimizing model performance and understanding the trade-offs involved in achieving accurate predictions.

Scope

The scope covers the following :

- Model Evaluation Metrics, Hyperparameter Optimization, Bias-Variance Trade-Off

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Students should demonstrate the ability to select and apply appropriate model evaluation metrics, interpret the results, and make informed decisions based on the performance of machine learning models.
- PC2.** Students' skills in optimizing model performance through hyperparameter tuning, with an emphasis on their ability to fine-tune model parameters effectively.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students should have a comprehensive understanding of various model evaluation metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and the ability to interpret and apply them in different machine learning scenarios.
- KU2.** Knowledge of hyperparameter tuning techniques, such as grid search, random search, and Bayesian optimization, is crucial. Students should understand how to adjust model settings for optimal performance.
- KU3.** Understanding the concept of the bias-variance trade-off, along with the implications of model complexity on accuracy and generalization, is important.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- SA1. Fill up documentation applicable to ones role.
 - SA2. Read and understand manuals, health and safety instructions, memos, other company documents.
 - SA3. Express statements or information clearly so that others can hear and understand.
 - SA4. Participate in and understand the main points of simple discussions.

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- GS2.**
- SB1. Follow organization rule-based decision making process.
 - SB2. Take decision with systematic course of actions and/or response.
 - SB3. Planning and organization of work to meet deadlines.
 - SB4. Work constructively and collaboratively with others.
 - SB5. Follow code of conduct.
 - SB6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
 - SB7. Recognize problems and search for solutions.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	75	26	20	-
PC1. Students should demonstrate the ability to select and apply appropriate model evaluation metrics, interpret the results, and make informed decisions based on the performance of machine learning models.	37	13	10	-
PC2. Students' skills in optimizing model performance through hyperparameter tuning, with an emphasis on their ability to fine-tune model parameters effectively.	38	13	10	-
NOS Total	75	26	20	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ITS/N1009
NOS Name	PERFORMANCE and ACCURACY of MACHINE LEARNING MODELS
Sector	Information Technology Sector
Sub-Sector	Artificial Intelligence
Occupation	AI Development
NSQF Level	4
Credits	4
Version	1.0
Next Review Date	NA

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise MCQ. Conduct of assessment is through trained proctors. Once the test begins, remote proctors have full access to the candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, the pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject experts including NIELIT officials.
5. The pass percentage would be 50% marks in each component.

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6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ITS/N1005.PROGRAMMING with PYTHON	28	17	-	-	45	12
NIE/ITS/N1006.CONCEPTUALISING DATA SCIENCE with PYTHON	28	16	10	-	54	15
NIE/ITS/N1007.DATA ANALYSIS and VISUALIZATION	42	25	-	-	67	19
NIE/ITS/N1008.FUNDAMENTALS of MACHINE LEARNING	25	8	-	-	33	10
NIE/ITS/N1009.PERFORMANCE and ACCURACY of MACHINE LEARNING MODELS	75	26	20	-	121	35
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	9
Total	218	122	30	-	370	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AI	Artificial Intelligence
AI	Artificial Intelligence
IDE	Integrated Development Environment
AI	Artificial Intelligence
IDE	Integrated Development Environment
AI	Artificial Intelligence
IDE	Integrated Development Environment

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.